
Fairness Auditing of an Income-Classification Model: SHAP-Based Analysis of Proxy Features

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Abstract

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1 Introduction

Machine-learning systems are increasingly used to support decisions that affect people’s opportunities, including hiring, credit, access to housing, and public services [Barocas et al., 2019, Corbett-Davies and Goel, 2017, Mehrabi et al., 2021]. In the trustworthy-AI literature, models are often shown to be accurate on average while still producing unequal error rates or prediction rates across demographic groups, particularly when historical inequities are encoded in the training data. These findings motivate fairness auditing as a requirement for disclosing how a model distributes outcomes and where prediction quality varies across groups.

A common first intervention is to remove a sensitive attribute such as *sex* from the model inputs. This is appealing because it is simple to explain and implement, but it does not guarantee that the remaining features are neutral with respect to the protected attribute. Other variables may still carry information associated with *sex*, either because they reflect social patterns or because they encode related aspects of an individual’s circumstances. We refer to such variables as potential *proxy features*. The central question of this study is therefore not only whether *sex* is present in a model, but whether the remaining feature set allows the model to recover information associated with *sex*.

To investigate this question, we use the UCI Adult Income dataset [Dua and Graff, 2019], a widely used benchmark containing demographic, educational, occupational, and work-related attributes with a binary target indicating whether annual income exceeds \$50,000. The dataset is useful for a controlled audit because it contains an explicit *sex* attribute and several variables that may encode social and economic structure. At the same time, its historical collection context and binary coding of *sex* limit what can be inferred from any single experiment. Our conclusions are therefore about proxy behavior in this audit design rather than a universal claim about fairness in income prediction.

Our study follows a reproducible comparison of model variants. We train a baseline XGBoost classifier with *sex* available and compare it with a model that excludes *sex*. We then examine *relationship*, *marital-status*, *occupation*, and *hours-per-week* as suspected proxies. TreeSHAP feature rankings, subgroup comparisons, and feature-ablation experiments are used as diagnostic tools for assessing whether these variables retain predictive information associated with the sensitive attribute. Figure 1 summarizes the overall audit pipeline and the comparison between the unconstrained and feature-restricted models. It proceeds from data preparation and paired model training through proxy identification, feature ablation, and fairness-constrained learning, followed by a common held-out evaluation of predictive performance and group fairness.

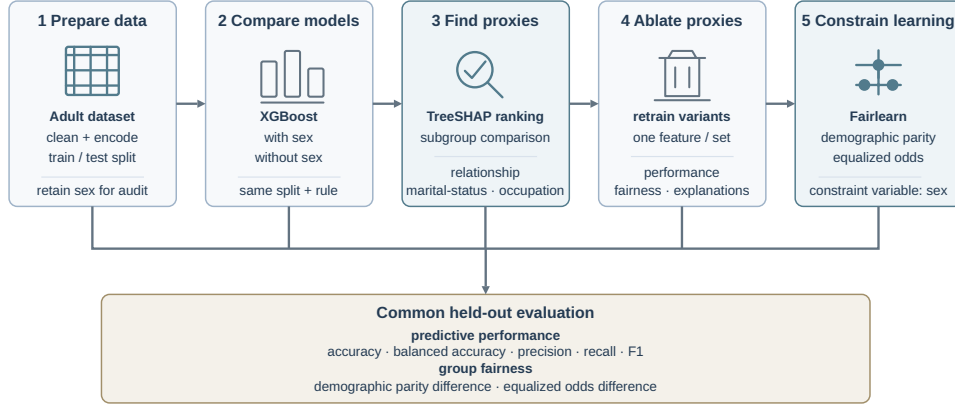


Figure 1: Overview of our proposed method. The audit compares explicit-feature removal, proxy ablation, and fairness-constrained learning under a common held-out evaluation.

This paper makes three practical contributions:

- It provides a reproducible audit of whether removing an explicit sensitive feature changes fairness measurements in an income-classification pipeline.
- It uses model explanations and ablation to test the narrower claim that particular remaining variables may act as proxies, rather than assuming that all correlated variables are interchangeable.
- It presents feature removal and algorithmic mitigation in a common comparison that distinguishes observed performance changes from the limitations of the selected fairness criteria.

The remainder of the paper is organized as follows. Section 2 ...

2 Related Work

Fairness criteria and trustworthy machine learning. A growing literature in trustworthy machine learning has emphasized that predictive accuracy alone does not determine whether a system is ethically or legally acceptable for decision-making [Barocas et al., 2019, Corbett-Davies and Goel, 2017, Mehrabi et al., 2021]. Researchers have formalized group-level criteria such as statistical parity, equal opportunity, and equalized odds, while also stressing that these criteria implement different notions of fairness and often conflict with one another in practice [Dwork et al., 2012, Hardt et al., 2016]. Our study follows this tradition by reporting both predictive performance and demographic-parity and equalized-odds differences rather than collapsing them into a single score.

Proxy features and explainability. The concern that protected-group information may be reconstructed from other features is central to work on proxy discrimination and fairness auditing [Datta et al., 2017]. Explainability methods provide a practical way to identify candidate variables that are highly predictive or strongly associated with subgroup differences [Lundberg and Lee, 2017]. We use TreeSHAP as a diagnostic lens to rank candidate proxy features. This is not a causal claim: a feature’s importance or subgroup dependence indicates potential proxy structure, not proof that the feature itself generates disparate treatment.

Mitigation strategies and subgroup analysis. A second line of work studies interventions at training time and at inference time. Constrained optimization and reductions-based fair classification show that it is possible to trade predictive performance against group-based constraints [Agarwal et al., 2018]. Related work on intersectional disparities also emphasizes that aggregate performance can conceal large subgroup differences [Buolamwini and Gebru, 2018]. Our audit situates feature

exclusion and fairness-constrained learning within a common evaluation setting, while keeping the dataset’s binary `sex` attribute and the associated limitations explicit.

3 Methodology

3.1 Our proposed method

We propose a reproducible, diagnostic audit protocol for testing whether removing an explicit sensitive attribute is sufficient to reduce group disparities. The protocol treats feature deletion as one intervention in a larger comparison rather than as evidence of fairness by itself. It has five stages: (i) construct a leakage-controlled preprocessing and evaluation pipeline, (ii) compare models with and without `sex`, (iii) identify candidate proxy features using explanations and subgroup diagnostics, (iv) measure the effect of ablating those candidates, and (v) compare feature-based interventions with constrained learning. The sensitive attribute is retained outside the model feature matrix throughout the experiment so that it can be used for subgroup evaluation.

Leakage-controlled pipeline. The supplied Adult training and test files are cleaned with one deterministic ETL procedure. Categorical variables are represented explicitly, including the documented missing-value category, and the income label is mapped to a binary target. Preprocessing components such as category encoding and numeric transformations are fitted on the training split only and then applied to the held-out test split. The target and `sex` are never used to fit preprocessing for the model features. This separation prevents information from the evaluation split from influencing training while preserving the attribute needed for fairness measurement.

Model comparison. The primary predictive model is an XGBoost classifier trained under a fixed configuration and evaluated with the same decision rule for every variant. We first fit a baseline using the full permitted feature set, including `sex`. We then fit a sex-excluded model after removing only `sex` from the feature matrix. This paired comparison isolates the effect of explicit attribute removal while holding the data split, preprocessing, model family, and evaluation procedure constant. Additional variants remove suspected proxies in a prespecified sequence, including `relationship`, `marital-status`, `occupation`, and `hours-per-week`. We report every attempted variant and do not interpret a feature’s removal as proof that the feature is intrinsically unfair.

Experimental setup. The baseline uses the official Adult training file (32,561 records) for fitting and the official Adult test file (16,281 records) for final evaluation; no additional random split is drawn. The model input contains the six numeric predictors and eight categorical predictors listed in the dataset, including `sex`. Numeric columns are passed through and categorical columns are one-hot encoded with unknown test categories ignored. The transformer is fitted on the training split only. The resulting baseline representation contains 108 transformed features. We use XGBoost with 300 trees, maximum depth 6, learning rate 0.1, histogram tree construction, and random seed 42; subsampling and column subsampling are both 1.0, and the training objective is binary logistic classification with log-loss evaluation. Predictions use the positive-class probability threshold 0.5. The saved test predictions retain `sex` and the observed income label solely for post-hoc evaluation.

Proxy diagnostics. Proxy analysis is performed only after fitting the sex-excluded model. We use TreeSHAP to rank feature contributions to individual predictions [Lundberg and Lee, 2017], summarize the rankings over the evaluation population, and compare feature distributions and prediction rates across the observed `sex` groups. Features are carried forward as candidate proxies when their model contribution and subgroup association provide convergent evidence in this dataset. We then perform feature ablation by retraining the model without one candidate or a prespecified candidate set. The ablation comparison distinguishes three quantities: predictive change, change in group metrics, and change in the explanatory profile. Because these diagnostics are observational and model-dependent, they identify candidate proxy pathways rather than establish causality.

Fairness-constrained comparison. As a complementary intervention, we train Fairlearn ExponentiatedGradient models with the broader feature representation and constraints based on demographic parity or equalized odds [Agarwal et al., 2018]. The sensitive attribute is supplied to the mitigation procedure as a constraint variable, not as an ordinary predictive feature. This comparison tests

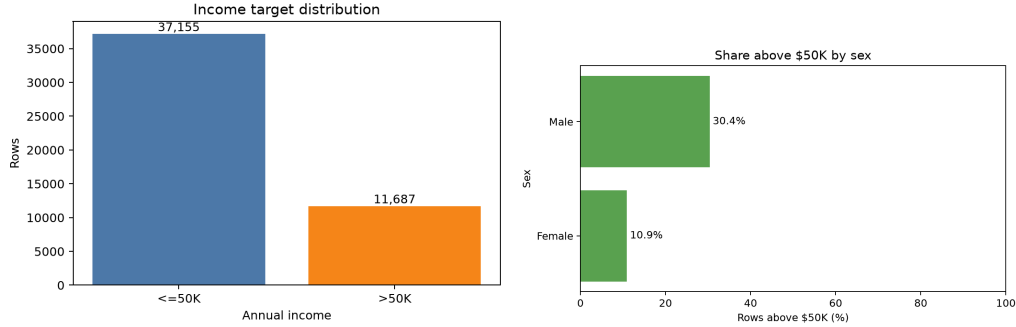


Figure 2: EDA of the cleaned Adult data. Left: target distribution. Right: observed proportion above \$50,000 by sex. These are descriptive dataset summaries, not predictions from a trained model.

whether an explicit group constraint changes the performance–fairness trade-off differently from deleting features. The unconstrained baseline, sex-excluded model, proxy-ablation variants, and constrained models are evaluated on the same held-out data and summarized in one comparison table.

Decision and reporting rule. For each variant, we report accuracy, balanced accuracy, precision, recall, F1, demographic-parity difference, and equalized-odds difference. Lower fairness-difference values indicate smaller observed disparities under the corresponding definition, but no single metric is treated as a sufficient fairness criterion. The protocol therefore reports predictive and fairness results separately, records feature sets and preprocessing decisions, and interprets changes only within the Adult dataset, selected model family, binary `sex` representation, and fixed prediction rule.

3.2 Dataset

We use the UCI Adult Income dataset [Dua and Graff, 2019], comprising 48,842 records and 14 predictors. The binary target is whether annual income exceeds \$50,000, and `sex` is retained as a sensitive attribute for evaluation. We preserve the dataset’s supplied training and test files rather than drawing a new split.

ETL The ETL implementation reads `adult.data` and `adult.test` from `data/adult`. It removes surrounding whitespace, removes the period appended to labels in the test file, validates the target labels and numeric fields, and converts the target to 0/1. The documented `?` values in categorical columns are mapped to an explicit `Missing` category rather than being silently dropped. This affected 6,465 values: 2,799 in `workclass`, 2,809 in `occupation`, and 857 in `native-country`. The cleaned training and test splits, combined dataset, and ETL metadata are written to `data/processed`.

EDA The exploratory analysis is descriptive and is performed on the cleaned combined dataset before model fitting. It reports descriptive statistics and creates visualizations for the target distribution, age by income class, income rates by education, numeric feature distributions, and documented missing-value counts. We also compare the target rate across the two observed `sex` categories as an initial descriptive fairness diagnostic; this comparison is not a model fairness result. In the cleaned data, 11,687 of 48,842 records are above \$50,000. The corresponding observed rates are 30.38% for males (9,918/32,650) and 10.93% for females (1,769/16,192).

3.3 Evaluation metrics

We evaluate each model on the held-out test split using both predictive performance and group fairness metrics. Predictive performance is summarized by accuracy, balanced accuracy, precision, recall, and F1 score. Accuracy is the fraction of correct predictions; balanced accuracy is the mean recall across the two income classes; precision is the fraction of predicted above-\$50,000 cases that are correct; recall is the fraction of actual above-\$50,000 cases that are identified; and F1 is the harmonic mean of precision and recall. Reporting both accuracy and balanced accuracy is useful because the target is not evenly distributed.

Table 1: Experimental data and preprocessing setup.

Item	Specification
Dataset	UCI Adult Income
Training/test rows	32,561 / 16,281
Predictors	14 (6 numeric, 8 categorical)
Target	Income above \$50,000 (binary)
Sensitive attribute	sex, retained for evaluation
Missing categorical values	6,465 mapped to Missing
Encoder	Training-fitted one-hot encoding
Classifier	XGBoost, 300 trees, depth 6, learning rate 0.1
Decision rule	Positive if predicted probability ≥ 0.5
Random seed	42

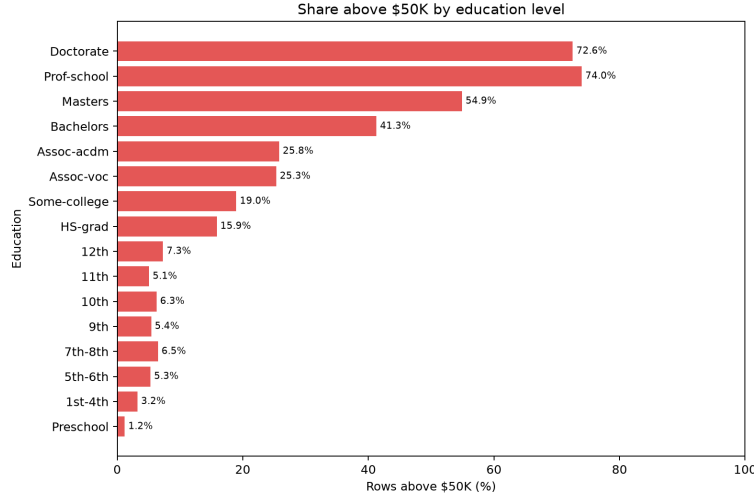


Figure 3: Observed proportion above \$50,000 by education category in the cleaned dataset.

Fairness metrics are computed across the observed binary `sex` categories, using the same fixed prediction rule for every model. The demographic-parity difference is defined as

$$\max_{g \in \mathcal{G}} P(\hat{Y} = 1 \mid A = g) - \min_{g \in \mathcal{G}} P(\hat{Y} = 1 \mid A = g),$$

where A is `sex` and $\mathcal{G}$ is the set of observed group labels. It measures disparity in positive prediction rates, without conditioning on the true income label. The equalized-odds difference is reported as the larger of the across-group differences in true-positive rates and false-positive rates:

$$\max \left\{ \max_g \text{TPR}_g - \min_g \text{TPR}_g, \max_g \text{FPR}_g - \min_g \text{FPR}_g \right\}.$$

Thus, it measures disparity in both types of conditional classification outcomes. Lower values indicate smaller observed group differences for both fairness measures, while zero does not establish that a system is fair in all relevant senses.

The comparison table will include the baseline XGBoost model with `sex`, a model with `sex` removed, models with suspected proxy features ablated, and Fairlearn ExponentiatedGradient models constrained by demographic parity or equalized odds. The same predictive and fairness metrics will be reported for each variant so that feature removal, proxy removal, and algorithmic mitigation can be compared without collapsing their different objectives into a single score. We do not perform robustness or threshold analyses; the reported values correspond to the prespecified evaluation procedure and fixed prediction rule.

Table 2: Descriptive statistics from the cleaned Adult dataset.

Group	Rows	Above \$50K	Rate (%)
Female	16,192	1,769	10.93
Male	32,650	9,918	30.38
All records	48,842	11,687	23.93

Table 3: Baseline XGBoost performance and fairness on the held-out test split ($n = 16,281$).

Metric	Value
Accuracy	0.8746
Balanced accuracy	0.7990
Precision	0.7786
Recall	0.6557
F1	0.7119
Demographic-parity difference	0.1710
Equalized-odds difference	0.0604

4 Results

4.1 Data preparation and descriptive audit

The ETL completed without dropping rows: all 48,842 records were retained, including the documented missing categorical values. The cleaned data contain 37,155 records at or below \$50,000 and 11,687 records above \$50,000. The descriptive target rate differs across the observed binary sex categories: 30.38% among males (9,918/32,650) and 10.93% among females (1,769/16,192). This is a property of the dataset and is not a prediction or a model fairness metric. It establishes the group labels and target distribution used in the subsequent held-out evaluation.

4.2 Baseline performance and fairness

Table 3 reports the first held-out baseline result. On the 16,281-record test split, the model achieved 87.46% accuracy and 79.90% balanced accuracy. Precision was 77.86%, recall was 65.57%, and F1 was 71.19%. These values describe the unconstrained model that includes sex; they are not evidence that the model is fair or unfair without reference to the explicitly reported group metrics.

The baseline positive prediction rate was 8.49% for females and 25.59% for males, producing a demographic-parity difference of 0.1710. The true-positive rates were 0.6102 and 0.6640, respectively, while the false-positive rates were 0.0207 and 0.0811. The equalized-odds difference was therefore 0.0604, driven by the false-positive-rate gap rather than the true-positive-rate gap. These results are descriptive of this fixed split, model, and threshold; they do not establish causal discrimination or generalize beyond the dataset and audit design.

5 Discussion

6 Conclusion

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Table 4: Baseline group-level rates on the held-out test split.

Sex	Rows	Selection rate	TPR	FPR
Female	5,421	0.0849	0.6102	0.0207
Male	10,860	0.2559	0.6640	0.0811

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